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SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEEING / TECHNOLOGY

MODEL QUESTION

ELECTRICAL MACHINE DESIGN

Maximum marks-100 Time: 3 hours

PART-A Marks

I. Answer the following questions in *one* or *two* sentences each question carries equal marks.

1. Define Rotating Hysteresis.
2. What is mean by leakage reactance of winding in transformer?
3. Define Chamfering in a DC machine
4. Why transposition of conductors required in a transformer.
5. What is the use of damper winding .

(2x5=10

PART-B

II. Answer any *five* of the following. Each question carries 6 marks.

1. List the important insulating materials used for transformers
2. The stator of a machine has a smooth surface but its rotor has open type of slots with slot width W_s= tooth width, W_t = 12 mm, and the length of air gap l_g =2mm. Find the effective length of air gap if the Carter's co-efficient = $1/(1+5\ l_g/W_s)$.
3. Differentiate between core type and shell type transformers.
4. Derive an output equation for a DC machine.
5. What are the effects of armature reaction? Explain each briefly.
6. Differentiate between squirrel cage motor and wound rotor motor.
7. Explain the constructional details of stator core of a Turbo alternator. (6x5=30

PART-C

UNIT-I

(Answer *one* full question from each unit. Each question carries 15 marks)

III. (a) Explain the significance of Real and Apparent flux densities in DC machine. 7

(b)Explain the significance of m.m.f distribution in a dc machine. 8

OR

IV What are important constraints considered foe specific electric loading 15

UNIT -II

V Determine the dimensions of core and yoke for a 200 kV, 50 Hz single phase core type transformer. A cruciform core is used with distance between adjacent limbs equal to 1.5 times the width of core laminations. Assume voltage per turn 12 V, maximum flux density 1.1 Wb/m^2, window space factor 0.32, current density 3 A / mm^2, and stacking factor is 0.9. The net iron area is 0.56 d^2 in a cruciform core where d is the diameter of circumscribing circle . Also the width of largest stamping is 0.85d. 15

OR

VI (a) What are the important constraints considered for Design of insulation for a transformer? Explain each briefly. 7

(b) Explain different type cooling system employed in transformer. 8

UNIT -III

VII (a) A design is required for a 50 kW, 4 pole ,600 rpm DC shunt generator , the full load terminal voltage being220V. If the maximum gap density is 0.83 Wb/m^2 and the armature ampere conductors per metre are 30000, calculate suitable dimensions of armature core to give a square pole face. Assume that the full load armature voltage drop is 3 % of the rated terminal voltage, and that the field current is 1% of rated full load current. Ratio of pole arc to pole pitch is 0.67 8

OR

VIII Explain different methods adopted for reducing armature reaction 15

UNIT –IV

IX (a) Explain the constructional details and speciality of 3 phase hydro generator stator core. 7

(b) Draw the different type stator slots for a 3 phase induction motor. 8

OR

X (a)Explain the method of Bracing of stator overhang in a synchronising machine 7

(b) design a 110 kW , 3300V , 3 phase,50 Hz, 600 synchronous rpm. Slip ring induction motor. The full load efficiency is 0.9 and the power factor is 0.86 8